# AUTOSAR High-Level Design (HLD) Document

## Subsystem: Integrated Vehicle Motion Control & V2X Communication | Version: 3.0

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## 1.0 System Architecture Overview & Scope (Revision 3.0)

This advanced High-Level Design (HLD v3.0) builds upon previous versions by integrating Vehicle-to-Everything (V2X) communication, Artificial Intelligence/Machine Learning (AI/ML) based decision making, and enhanced cybersecurity measures. The architecture now supports cooperative adaptive cruise control (CACC), platooning, and over-the-air (OTA) update capabilities.

## 1.1 Updated Software Components Definition

• **BrakeController**: Application SWC (ASIL-D). Now includes predictive braking based on V2X inputs and AI-driven road condition analysis.
• **VehicleController**: Application SWC (ASIL-D). Expanded to manage V2X communication, AI/ML model inference, and cooperative vehicle control.
• **SensorFusion**: Application SWC (ASIL-D). Now incorporates V2X perception data and AI-based object classification with confidence scoring.
• **RadarInterface**: Sensor SWC (ASIL-B). Enhanced with AI-based target tracking and micro-Doppler signature analysis.
• **CameraInterface**: Sensor SWC (ASIL-B). Now includes neural network inference for semantic segmentation and 3D object detection.
• **ADASCruiseControl**: Application SWC (ASIL-C). Enhanced with cooperative adaptive cruise control (CACC) and platooning capabilities.
• **AIDecisionMaker**: [NEW] Application SWC (ASIL-D). Processes sensor and V2X data using deep learning models for trajectory planning and risk assessment.
• **V2XCommunicationManager**: [NEW] Communication SWC (ASIL-B). Handles DSRC/C-V2X messaging, security credentials, and message validation.
• **CybersecurityManager**: [NEW] Complex SWC (ASIL-D). Implements intrusion detection, secure boot, and runtime application self-protection (RASP).
• **OTAUpdateManager**: [NEW] Application SWC (ASIL-B). Manages secure over-the-air software updates and version control.
• **SafetyController**: Complex SWC (ASIL-D). Now includes runtime monitoring of AI/ML models for functional safety compliance.
• *(Note: Legacy CAN_Gateway component removed in favor of unified Ethernet/V2X communication stack.)*

## 2.0 Updated Software Component Port Specifications

| Component (SWC) | Type | ASIL | Provided Ports (P-Port) | Required Ports (R-Port) |
|---|---|---|---|---|
| BrakeController | Application SWC | ASIL-D | P_BrakeCommand, P_BrakeStatus, P_WheelSpeed | R_VehState, R_ADAS_DecelReq, R_V2X_Hazar |
| VehicleController | Application SWC | ASIL-D | P_VehState, P_TargetTorque, P_V2X_StatusBroadcast | R_BrakeCommand, R_AI_PlanningStatus, R_ADAS_Moti |
| SensorFusion | Application SWC | ASIL-D | P_FusionTrack, P_ObstacleDist, P_LaneGeometry | R_RadarRaw, R_CameraVisionRaw, R_V2X_RemoteSensorData, R_V2X_Rem |
| RadarInterface | Sensor SWC | ASIL-B | P_RadarRaw | R_EthData_Radar, R_V2X_RadarConfig |
| CameraInterface | Sensor SWC | ASIL-B | P_CameraVisionRaw | R_EthData_Camera, R_V2X_CameraConfig |
| ADASCruiseControl | Application SWC | ASIL-C | P_ADAS_DecelReq, P_ADAS_MotionRequest | R_FusionTrack, R_VehState, R_V2X_PlatoonSta |
| AIDecisionMaker | Application SWC | ASIL-D | P_AI_RiskAssessment, P_AI_PlanningStatus | R_SensorFusionTrack, R_V2X_AI_Input, R_SafetyM |
| V2XCommunicationManager | Communication SWC | ASIL-B | P_V2X_MessageTx, P_V2X_Status, P_V2X_SecurityEvent | R_V2X_MessageRx, R_V2X_ConfigUpdate, R_C |
| CybersecurityManager | Complex SWC | ASIL-D | P_SecurityAlert, P_SecureBootStatus | R_V2X_SecurityEvent, R_OTA_Verification, R_Sy |
| OTAUpdateManager | Application SWC | ASIL-B | P_UpdateStatus, P_VerificationResult | R_UpdateRequest, R_VerificationChallenge, R_C |
| SafetyController | Complex SWC | ASIL-D | P_SafeStateCmd, P_AI_ModelStatus | R_BrakeCommand, R_VehState, R_ADAS_Dece |

## 3.0 Updated Interfaces & Signal Routing

• **ADAS_DecelRequestInterface**: Source **ADASCruiseControl** -> Target **BrakeController**. Signals: TargetDecelerationReq (float32), UrgencyLevel (uint8), AEB_Arm (boolean).
• **BrakeCommandInterface**: Source **BrakeController** -> Target **VehicleController**. Signals: TargetDeceleration (float32), BrakePedalStatus (uint8), EBD_Active (boolean), RegBrakeTorque (float32).
• **CameraVisionStreamInterface**: Source **CameraInterface** -> Target **SensorFusion**. Signals: ObjectBoundingBoxes, LaneCurvature, PedestrianDetectionFlag.
• **SensorTelemetryInterface**: Source **SensorFusion** -> Target **ADASCruiseControl**. Signals: LeadVehicleDist, LeadVehicleSpeed, TargetCutInProb.
• **V2X_HazardAlertInterface**: [NEW] Source **V2XCommunicationManager** -> Target **BrakeController**. Signals: HazardType (enum), Distance (float32), TimeToCollision (float32), Reliability (uint8).
• **AI_RiskAssessmentInterface**: [NEW] Source **AIDecisionMaker** -> Target **BrakeController**. Signals: RiskLevel (float32), RecommendedAction (enum), Confidence (float32).
• **AI_PlanningInterface**: [NEW] Source **AIDecisionMaker** -> Target **VehicleController**. Signals: TrajectoryPoints (array), VelocityProfile (array), SafetyMargin (float32).
• **V2X_PlatoonStatusInterface**: [NEW] Source **V2XCommunicationManager** -> Target **ADASCruiseControl**. Signals: PlatoonID (uint16), PositionInPlatoon (uint8), SpacingDistance (float32).
• **CyberSecurityAlertInterface**: [NEW] Source **CybersecurityManager** -> Target **SafetyController**. Signals: ThreatLevel (enum), AffectedComponent (string), RecommendedAction (enum).
• **AI_ModelMonitorInterface**: [NEW] Source **SafetyController** -> Target **AIDecisionMaker**. Signals: ModelHealthStatus (enum), DeviationMetrics (struct), ResetRequired (boolean).

## 4.0 Functional Safety & Security Considerations (ISO 26262 & ISO/SAE 21434)

Safety Goal SG-01: 'Prevent unintended deceleration or total loss of braking' remains ASIL-D. Additional safety goals include SG-02: 'Prevent unsafe AI/ML behavior' (ASIL-D) and SG-03: 'Maintain secure vehicle communication' (ASIL-B). The BrakeController, VehicleController, AIDecisionMaker, and SafetyController SWCs are allocated ASIL-D development lifecycle rigor. SensorFusion is assigned ASIL-D due to its critical role in AI/ML perception pipeline. V2XCommunicationManager and CybersecurityManager address ISO/SAE 21434 cybersecurity requirements with ASIL-B and ASIL-D respectively.

## 5.0 Execution Schedule & Communication Latency Budgets

• **CameraInterface_Step (30ms Periodic)**: Processes vision frame and runs neural network inference.
• **RadarInterface_Step (20ms Periodic)**: Processes radar data with CFAR detection and AI-based tracking.
• **SensorFusion_Step (20ms Periodic)**: Multi-modal fusion including V2X perception data.
• **AIDecisionMaker_Step (50ms Periodic)**: Runs deep learning models for risk assessment and trajectory planning.
• **V2XCommunicationManager_Step (100ms Periodic)**: Handles message transmission, reception, and security processing.
• **CybersecurityManager_Step (500ms Periodic)**: Performs security monitoring and intrusion detection checks.
• **BrakeController_Step (10ms Periodic)**: Closed-loop deceleration actuation with predictive braking inputs.